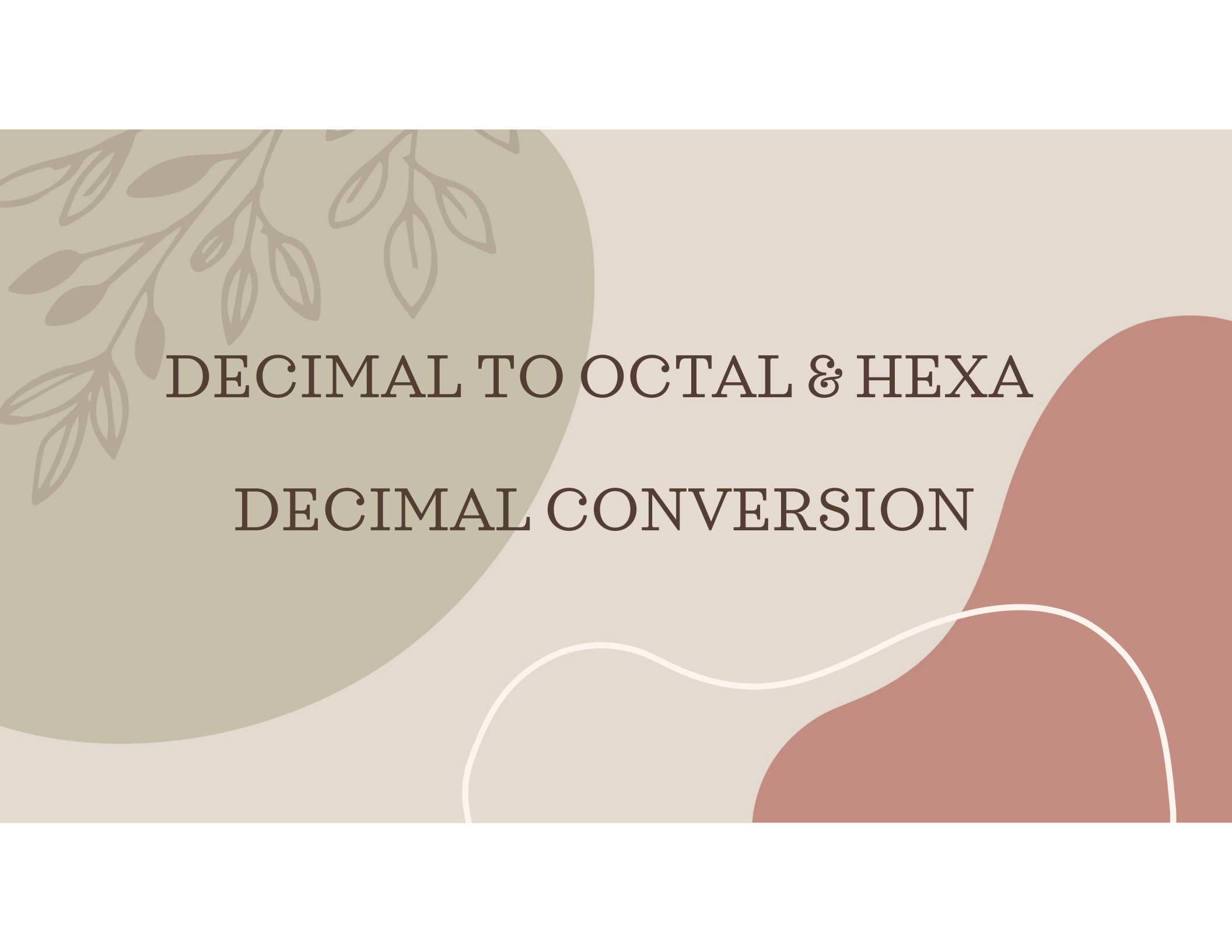




DECIMAL TO OCTAL & HEXA DECIMAL CONVERSION

3-Bit Representation

- A 3-bit binary number consists of 3 binary digits.
- The total number of combinations possible is 8
- Therefore, a 3-bit system can represent decimal numbers from 0 to 7.

3-Bit Binary Representation

Decimal	3-Bit Binary
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

4-Bit Representation

- Similarly, a 4-bit binary number consists of 4 binary digits.
- The total number of combinations is $2^4 = 16$
- Therefore, a 4-bit system can represent decimal numbers from 0 to 15.

10	1010
11	1011
12	1100
13	1101
14	1110
15	1111

A = 10
B = 11
C = 12
D = 13
E = 14
F = 15

4-Bit Binary Representation

Decimal	4-Bit Binary
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

Decimal to Octal Conversion

Octal is a base-8 number system.

0, 1, 2, 3, 4, 5, 6, and 7

- repeatedly divide the integer part by 8 and write down the remainders
- multiply the fractional part by 8
- The octal equivalent is obtained by reading the remainders from bottom to top.

Example 1

$$112_{10} = (\quad)_8$$

$$112_{10} = (160)_8$$

Handwritten conversion of 112 to octal:

$$\begin{array}{r} 8 \overline{) 112} - 0 \\ 8 \overline{) 14} - 6 \\ 1 \end{array}$$

Remainders: 0, 6, 1 (read from bottom to top)

$$(112)_{10} \rightarrow (160)_8$$

Example 2 $25.625_{10} = (\quad)_8$

Handwritten conversion of 25.625 to octal:

$$2. (25.625)_{10} \rightarrow (\quad)_8$$
$$\begin{array}{r} 8 \overline{) 25} - 1 \\ 8 \overline{) 3} \end{array} \Rightarrow (31)_8$$
$$\Rightarrow (31.50)_8$$

$$25.625_{10} = (31.50)_8$$

Example 3 $2048_{10} = (\quad)_8$

$$2048_{10} = (3800)_8$$

Handwritten conversion of 2048 to octal:

$$(i) (2048)_{10} \rightarrow (\quad)_8$$
$$\begin{array}{r} 8 \overline{) 2048} - 0 \\ 8 \overline{) 248} - 0 \\ 8 \overline{) 31} - 7 \\ 3 \end{array}$$

Remainders: 0, 0, 7, 3 (read from bottom to top)

$$(2048)_{10} \rightarrow (3800)_8$$

Decimal to Hexadecimal Conversion

Hexadecimal is a base-16 number system.

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15

A = 10 B = 11 C = 12 D = 13 E = 14 F = 15

- repeatedly divide the integer part by 16 and write down the remainders

multiply the fractional part by 16

- The hexadecimal equivalent is obtained by reading the remainders from bottom to top.

Example 1

$$26_{10} = ()_{16}$$

$$26_{10} = 1A_{16}$$

Handwritten calculation showing the conversion of 26 to hexadecimal. It shows 16 dividing 26 with a remainder of 10. The remainder 10 is then converted to the hexadecimal digit A. The final result is 1A.

$$\begin{array}{r} 16 \overline{) 26} - 10 \\ \hline \Rightarrow 1 \ 10 \\ \Rightarrow 1 \ A \\ \hline \end{array}$$

Example 2 $255_{10} = ()_{16}$

$$255_{10} = FF_{16}$$

Handwritten calculation showing the conversion of 255 to hexadecimal. It shows 16 dividing 255 with a remainder of 15. Then 16 divides 15 with a remainder of 15. The remainders 15 and 15 are converted to the hexadecimal digits F and F respectively. The final result is FF.

$$\begin{array}{r} 16 \overline{) 255} - 15 \\ 16 \overline{) 15} - 15 \\ \hline 0 \\ \hline \end{array}$$

0 15 15
 $\Rightarrow FF$

Example 3 $100_{10} = ()_{16}$

$$100_{10} = 64_{16}$$

Handwritten calculation showing the conversion of 100 to hexadecimal. It shows 16 dividing 100 with a remainder of 4. Then 16 divides 6 with a remainder of 6. The remainders 4 and 6 are converted to the hexadecimal digits 4 and 6 respectively. The final result is 64.

$$\begin{array}{r} 16 \overline{) 100} - 4 \\ 16 \overline{) 6} - 6 \\ \hline 0 \\ \hline \end{array}$$

0 64 $\Rightarrow (64)_{16}$

General Rule for Decimal to Any Number System Conversion

1. For the Integer Part

- Repeatedly **divide** the integer part by the **radix (base)** of the destination number system.
- Note down the remainders.
- Read the remainders from **bottom to top** to get the converted integer part.

2. For the Fractional Part

- Repeatedly **multiply** the fractional part by the **radix (base)** of the destination number system.
- Note down the integer part obtained after each multiplication.
- Read the integer parts from **top to bottom** to get the converted fractional part.

Practice Questions

1. Convert 18_{10} to octal.
2. Convert 125_{10} to octal.
3. Convert 125_{10} to hexadecimal.
4. Convert 150_{10} to hexadecimal.
5. Convert 2025_{10} to octal.
6. Convert 2025_{10} to hexadecimal.
7. Convert 0.625_{10} to binary.
8. Convert 0.75_{10} to octal.
9. Convert 0.5_{10} to hexadecimal.
10. Convert 12.375_{10} to binary.
11. Convert 25.625_{10} to octal.



thank you

Introduction to number systems
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